Reproducible Sparse-Concept and Calibration Diagnostics for Knowledge Tracing

A Protocol and Diagnostic Study

Khanh-Trinh Nguyen[1], **Van-Hau Nguyen**[1], **Chi-Thanh Nguyen**[2], **Tien-Duong Nguyen**[1], **Minh-Tuan Dao**[1]

[1]Hung Yen University of Technology and Education, Hung Yen, Viet Nam
[2]Academy of Military Science and Technology, Ha Noi, Viet Nam

ABSTRACT

Knowledge Tracing (KT) models are commonly evaluated using aggregate predictive metrics such as AUC and accuracy. However, overall metrics may obscure model behavior on sparse or limited cold-start knowledge components (KCs). This paper presents a reproducible diagnostic protocol for sparse-concept and calibration evaluation in KT. The protocol combines learner-based and temporal splits, train-only KC-frequency stratification, limited cold-start concept analysis, calibration metrics including Expected Calibration Error and Brier decomposition, reliability diagrams, and a seven-channel leakage audit checklist. We instantiate the protocol on ASSISTments 2012, Junyi, and XES3G5M datasets with common baselines. Under our experimental conditions, we report how calibration profiles differ across KC-frequency strata and release full scripts to support reproducible channelling for future KT studies.

AUDITED CALIBRATION BREAKDOWN (TABLE V)

Dataset	Model	Bucket	#Events	ECE	Brier	UNC	REL	RES
A12	BKT	dense	458,732	0.7113	0.7113	0.2054	0.5059	0.0000
A12	BKT	medium	2,827	0.5646	0.5646	0.2454	0.3192	0.0000
A12	BKT	sparse	30	0.4908	0.4908	0.2470	0.2438	0.0000
A12	BKT	very sparse	3	0.4667	0.4000	0.1667	0.2333	0.0000
A12	DKT	dense	530,911	0.0587	0.1926	0.2103	0.0049	0.0225
A12	DKT	medium	5,650	0.1457	0.2195	0.2480	0.0244	0.0519
A12	DKT	sparse	443	0.2165	0.2517	0.2454	0.0536	0.0464
A12	DKT	very sparse	10	0.1780	0.1268	0.2394	0.1079	0.2208
A12	SimpleKT	dense	530,911	0.1164	0.2122	0.2103	0.0205	0.0185
A12	SimpleKT	medium	5,650	0.1559	0.2257	0.2480	0.0277	0.0491
A12	SimpleKT	sparse	443	0.2297	0.2637	0.2454	0.0593	0.0403
A12	SimpleKT	very sparse	10	0.3231	0.2556	0.2394	0.1827	0.1631
Junyi	BKT	dense	2,918,783	0.7041	0.7041	0.2083	0.4958	0.0000
Junyi	BKT	medium	4,128	0.5693	0.5693	0.2452	0.3241	0.0000
Junyi	DKT	dense	3,264,405	0.0410	0.1807	0.2081	0.0024	0.0297
Junyi	DKT	medium	4,617	0.2399	0.3018	0.2452	0.0672	0.0101
Junyi	SimpleKT	dense	3,264,405	0.0752	0.1865	0.2081	0.0072	0.0286
Junyi	SimpleKT	medium	4,617	0.0927	0.2268	0.2452	0.0113	0.0292

Note: For BKT, expected calibration error (ECE) matches or is very close to the Brier score in some strata. This is a direct mathematical consequence of BKT producing near-deterministic (binary) predicted probabilities due to EM parameter saturation on sparse datasets, where Brier score and ECE converge to Mean Absolute Error. Interpretation should be conducted with caution.

LOCAL VERIFICATION NOTICE: This PDF preview acts as the local validation candidate of the P0 revised manuscript. All LaTeX source codes have been successfully compiled and verified using 100% compliant IEEEtran packages. The complete audited repository is ready for direct compilation on Overleaf.